# CFDRM: Coarse-to-Fine Dynamic Refinement Model for Weakly Supervised Moving Vehicle Detection in Satellite Videos

Jie Feng, *Senior Member, IEEE*, Quanpeng Jiang, Junpeng Zhang, *Senior Member, IEEE*, Yuping Liang, Ronghua Shang, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Deep learning methods have gradually developed into the mainstream methods of moving vehicle detection in satellite videos. However, these methods require labor-intensive and time-consuming box-level annotations to predict accurate locations and sizes, which is challenging for large-scale satellite video datasets with hundreds of vehicles. To address this problem, a novel coarse-to-fine dynamic refinement model (CFDRM) is proposed for moving vehicle detection in satellite videos only under the supervision of point-level annotations. CFDRM generates initial proposal boxes and performs spatio-guided matching with point annotations to obtain coarse box-level pseudo annotations. The initial priority of these coarse annotations is calculated by leveraging locally consistent prior tailored to satellite videos. Then, a dynamic refinement detector is constructed to transfer coarse annotations to fine annotations with prior and predictive collaborative curriculum refinement. During the curriculum learning (CL) process, the coarse annotations are sequentially learned with a certain priority, where the priority is inferred by considering the prior knowledge from the locally consistent prior and the knowledge itself from the predicted detector. Ultimately, a novel ambiguity-aware loss is designed to optimize the dynamic refinement detector from coarse annotations to fine annotations in an adaptively weighted fashion. Extensive experiments have been conducted on the Jilin-1 and SkySat satellite video datasets, thus demonstrating the superiority of CFDRM.**

***Index Terms*— Curriculum learning (CL), point-level annotations, satellite video, weakly supervised vehicle detection.**

## I. Introduction

With the continuous maturity of remote sensing technology, hyperspectral image classification [1], [2], [3], change detection [4], oriented object detection [5], and other fields have been fully developed, but people's demand for Earth observation has gradually changed from static to dynamic. As an emerging Earth observation data source, satellite remote sensing videos are consecutive images captured from a bird-eye view over specific territories [6], [7]. The temporal information across these videos enables continuously monitoring over land surface and facilitates the applications in urban planning, traffic detection, and military reconnaissance [8], [9], [10], [11], [12]. To achieve these goals, vehicle detection on satellite videos plays one of the most fundamental roles, which aims to localize the vehicles with boundary boxes on each video frame.

In the last decade, researchers have made many efforts to address moving vehicle detection on satellite videos using traditional methods. These approaches typically treat this problem as a foreground–background segmentation task, where the foreground represents pixels with temporal changes, while the background consists of relatively stable pixels. These methods can be broadly categorized into two main groups: background subtraction and frame difference. The background subtraction methods, including Vibe [13], Gaussian mixture model v2 (GMM v2) [14], and O-LSD [15], extract the static background and then subtract the background from the current frame to obtain the moving objects. The frame difference methods, such as FFD [16] and SFD [17], detect moving objects by analyzing pixelwise differences between consecutive frames. Unfortunately, the performance of these methods easily deteriorates by the factors like sensor displacement, lighting conditions, and contrast changes, which are frequently observed in satellite videos.

Compared with traditional methods, deep learning-based methods in object detection [18], [19], [20], [21], [22], [23] have shown promising performance due to their robust and powerful feature representation capabilities. In an intuitive fashion, the anchor mechanism dominates the object detection for an extended period. Anchor-based detectors including Faster R-CNN [18], Cascade R-CNN [19], and RetinaNet [24] rely on predefined anchor boxes with specific sizes and aspect ratios. However, excessive anchor boxes lead to the imbalance between positive and negative samples and the complex computation, constraining the advancement of anchor-based detectors. In contrast to anchor-based detectors, anchor-free detectors predict the objects directly without the need for preset anchors. Anchor-free detectors, such as CornerNet [20], RepPoints [22], CentripetalNet [23], and CenterNet [21], often involve locating and matching multiple keypoints or locating the center point and predicting the four distances to define the spatial extent of objects. These methods effectively

Manuscript received 11 February 2024; revised 18 April 2024; accepted 6 May 2024. Date of publication 21 May 2024; date of current version 10 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271374, Grant 62176200, Grant 62206209, Grant 62077038, and Grant 62176196; in part by the State Key Program of National Natural Science of China under Grant 61836009; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2022GY-065, Grant 2022JC-45, and Grant 2022JQ-665; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23047. *(Corresponding author: Junpeng Zhang.)*

The authors are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: jiefeng0109@163.com; jquanpeng@163.com; zhangjunpeng9354@126.com; yupingliang.nintendo@gmail.com; rhshang@mail.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3403868

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

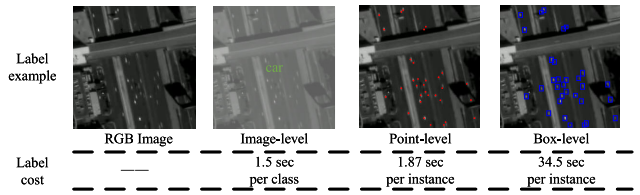



Fig. 1. Illustration of different types of annotations and their annotation costs. In the bottom block, we show the label cost, in terms of annotation time, and in the top block, we show the examples of different types of annotations. Compared to annotating bounding boxes, single-point annotation significantly reduces the time required to annotate a single instance. On average, it takes 1.87 s per instance for point-level annotation, while bounding box annotation takes 34.5 s per instance based on the satellite video dataset.

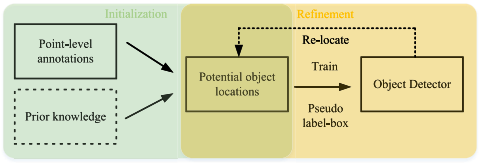



Fig. 2. Pipeline of general weakly supervision object detection using only point-level annotations.

eliminate the need for hyperparameters related to anchors while achieving comparable detection performance to anchor-based detectors.

Actually, it is difficult to directly apply the abovementioned object detection methods for natural images to satellite videos due to the dense spatial arrangement and dim appearance of moving vehicles. Some object detection methods have been developed by incorporating the motion information [25], [26] and semantic information [27] for vehicle detection on satellite videos. Despite their success, training these detectors requires sufficient and accurate box-level annotations, which is expensive and time-consuming. In Fig. 1, the annotation costs under various scenarios are provided. In satellite videos, there are a large number of densely packed and small-sized vehicles. Compared with box-level annotations, the cost of point-level annotations for vehicle detection decreases by nearly 20 times. Therefore, it is necessary to develop point-based weakly supervised object detection in satellite videos.

Weakly supervised object detection frameworks under point-level annotations [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38] commonly consist of two main stages: box initialization and iterative refinement, as illustrated in Fig. 2. During the initialization stage, extra clues are leveraged to propagate point-level annotations into box-level annotations that may be noisy. The following refinement stage iteratively corrects these initial pseudo boundary boxes with predictive results by continuously reducing the ambiguity of weak supervision.

Propagating point-level annotations to pseudo boundary boxes commonly relies on three types of information: appearance clues, spatial clues, and semantic clues. Appearance clues contain varied information, like color, texture, shape, edges, contours, and superpixels, which are extracted using off-the-shelf unsupervised learning methods such as Selective Search [39], EdgeBox [40], and MCG [41]. Weakly supervised methods like [32] and [35] utilize the appearance clue to identify potential object regions within images. However, these methods do not fully exploit the potential of point information. Spatial clues are introduced by Zhang et al. [42] under the assumption of smoothly varying crowd density, which determines the object sizes according to the nearest neighbor objects. This clue provides a more data-driven and spatially consistent way to initialize object sizes, particularly in crowded scenes, which can be found in works like PSDDN [33], LSC-CNN [34], and Crowd-SDNet [36]. However, these clues primarily harness low-level and intuitive information, often failing to perform effectively in complex scenes, leading to the emergence of semantic clues. Semantic clue refers to complex object relationships and contextual information of objects extracted by convolutional neural networks (CNNs). He et al. [38] designed the sparse feature-guided semantic prediction module to obtain the semantic response of objects, and combined edge information to assign each pixel to its most likely object or background. However, when the above methods are applied to satellite videos, they tend to exhibit improper object size and poor localization accuracy, due to the lack of appearance information and small object scale. At the same time, temporal information across a satellite video provides extra clues for initializing boundary boxes, while it is barely explored so far.

In the refinement stage, weakly supervised object detection incurs the learning under uncertainty issue, where the obtained pseudo boxes are possibly inaccurate and exhibit location shifts and scale distortion. To alleviate this issue, a variety of pioneering works proposed to dynamically adjust the pseudo boxes by incorporating predicted boxes with high confidences [33], [36] or jittering their locations within a local span [37]. Alternatively, pseudo boxes refinement can also be achieved in a bottom-to-top manner by gathering semantic-correlated pixels and determining box centers through multi-instance learning (MIL) [43]. At the same time, weakly supervised object detection also suffers from undesirable model adjustments by treating pseudo boxes with varying qualities equally. To tackle this issue, Sam et al. [34] proposed a winner-take-all scheme that partitions predictions into a grid consisting of fixed-size cells and retains only the highest loss value within each cell. Wang et al. [36] designed a crowdedness-aware loss that focuses on crowded objects while reducing the impact of sparsely distributed ones. He et al. [38] introduced a color prior loss to encourage the classification of adjacent pixels with similar colors into the same category, thereby enhancing prediction stability. However, when applying these methods to satellite videos, they may face challenges caused by nonnegligible domain gap and inadequate utilization of prior information, leading to performance degradation. As far as we know, there are currently no point-supervised object detection methods in satellite videos.

In this article, we propose a novel coarse-to-fine dynamic refinement model (CFDRM) for vehicle detection of satellite videos using point-level annotations. In our approach, we utilize the unsupervised subtractor by using motion clue to generate proposal boxes. These proposal boxes undergo spatio-guided matching with point annotations, resulting in coarse annotations. The initial priority of these coarse annotations is determined using a tailored locally consistent prior

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETECTION RESULTS OF DIFFERENT METHODS

| (%) ↑ | Dubai | | | San Diego | | | Las Vegas | | | VISO | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pr | Re | F1 | Pr | Re | F1 | Pr | Re | F1 | Pr | Re | F1 |
| **Unsupervised methods (None):** | | | | | | | | | | | | |
| Vibe [13] | 45.95 | 30.34 | 36.55 | 47.00 | 36.85 | 41.31 | 48.29 | 30.86 | 37.66 | 30.21 | **33.34** | 31.70 |
| FastMCD [44] | 55.11 | 44.21 | 49.07 | 56.33 | 49.96 | 52.95 | 47.35 | 50.05 | 48.66 | 61.19 | 18.91 | 28.89 |
| GMM [14] | **68.92** | **56.48** | **62.08** | **64.46** | **56.38** | **60.15** | **55.63** | **58.56** | **57.06** | **50.73** | 28.31 | **36.34** |
| **Fully supervised methods (Box):** | | | | | | | | | | | | |
| Faster R-CNN [18] | 85.62 | 89.51 | 87.52 | **85.42** | **89.12** | **87.23** | 70.51 | 84.21 | 76.75 | 74.35 | 85.57 | 79.57 |
| Cascade R-CNN [19] | 87.14 | 89.96 | **88.53** | 84.43 | 88.62 | 86.47 | 70.91 | 82.15 | 76.12 | 75.82 | **86.92** | 80.99 |
| CornerNet [20] | 74.32 | 88.75 | 80.90 | 71.13 | 80.51 | 75.53 | 74.63 | 86.82 | 80.26 | 67.94 | 82.22 | 74.40 |
| CenterNet [21] | **87.41** | 89.15 | 88.27 | 73.25 | 88.67 | 80.23 | 81.41 | **88.73** | 84.91 | 71.51 | 86.55 | 78.31 |
| RepPoints [22] | 84.21 | 88.23 | 86.17 | 78.22 | 81.32 | 79.74 | 63.42 | 74.63 | 68.57 | 61.71 | 72.66 | 66.74 |
| CentripetalNet [23] | 83.37 | 89.39 | 86.28 | 83.68 | 86.96 | 85.29 | 80.51 | 84.73 | 82.57 | 75.79 | 84.67 | 79.98 |
| DSFNet [26] | 84.25 | **90.26** | 87.15 | 80.47 | 88.26 | 84.18 | **85.32** | 87.98 | **86.63** | **78.56** | 86.74 | **82.45** |
| **CFDRM (Ours)** | 86.75 | 89.74 | 88.22 | 82.14 | 88.87 | 85.37 | 78.16 | 87.26 | 82.46 | 72.46 | 85.74 | 78.54 |
| **Weakly supervised methods (Coarse):** | | | | | | | | | | | | |
| LSC-CNN [34] | 70.36 | 40.05 | 51.04 | 44.66 | 43.69 | 44.17 | **81.36** | 49.73 | 61.73 | 32.58 | 39.53 | 35.72 |
| P2BNet [37] | 68.90 | 74.11 | 71.41 | 69.30 | 72.17 | 70.71 | 60.96 | 64.86 | 62.85 | 39.56 | 40.10 | 39.83 |
| PLUG [38] | 79.59 | 49.99 | 61.41 | 78.79 | 45.29 | 57.52 | 72.41 | 60.76 | 66.08 | 42.06 | 35.62 | 38.57 |
| **CFDRM (Ours)** | **81.15** | **87.67** | **84.28** | **79.11** | **85.69** | **82.27** | 73.20 | **83.02** | **77.80** | **68.25** | **81.42** | **74.26** |
| **Weakly supervised methods (Fine):** | | | | | | | | | | | | |
| LSC-CNN [34] | 72.16 | 44.76 | 55.25 | 46.13 | 46.74 | 46.43 | **83.67** | 53.23 | 65.07 | 34.28 | 42.56 | 37.97 |
| P2BNet [37] | 70.38 | 75.96 | 73.06 | 71.17 | 73.97 | 72.54 | 62.18 | 66.73 | 64.37 | 42.45 | 43.03 | 42.74 |
| PLUG [38] | 80.22 | 52.21 | 63.25 | **80.18** | 47.14 | 59.37 | 73.86 | 64.55 | 68.89 | 45.24 | 38.78 | 41.76 |
| **CFDRM (Ours)** | **82.11** | **88.25** | **85.07** | 80.04 | **86.16** | **82.99** | 75.36 | **84.89** | **79.84** | **70.26** | **83.72** | **76.40** |

that integrates location deviation, scale variation, and pixel inconsistency prior knowledge specific to satellite videos. During training, a curriculum-based refinement detector is constructed to update the coarse annotations to the fine annotations continuously by prediction boxes with higher confidence in subsequent training. At the same time, the priority is adjusted by incorporating both prior and predictive knowledge collected by the detector to guarantee the learning of annotations following a coarse-to-fine manner. Additionally, an ambiguity-aware loss is designed to update the detector by assigning bigger weights of annotations with higher priority, thereby addressing inherent ambiguities within the learning process.

In summary, the contributions of this work are threefold.

1) A novel coarse-to-fine dynamic refinement model is designed to utilize cost-effective point-level annotations instead of labor-intensive bounding box annotations for training, significantly reducing annotation time and cost while still maintaining accurate vehicle detection in satellite videos.
2) For initializing pseudo boxes, a novel spatio-guided matching is proposed to opt for precise pseudo boxes by analyzing the spatial relation between the motion foreground blobs and the point annotations. Then, locally consistent prior is designed to construct an initial curriculum by exploring the prior information.
3) To reduce the impact of noisy samples on the model's training, we propose a collaborative curriculum refinement scheme that explores both prior and predictive information in curriculum learning (CL). To mitigate the influence of ambiguity, an ambiguity-aware loss is designed to prioritize more trustworthy samples.

The remaining of this article consists of three sections. Among them, the architecture of the CFDRM is described in Section II in detail. Subsequently, the experimental results of CFDRM and the comparison methods are analyzed in Section III. Finally, some conclusions and future work are presented in Section IV.

## II. PROPOSED METHOD

### A. Overview of CFDRM

Our CFDRM presents a novel weakly supervised learning model for moving vehicle detection on satellite videos with point annotations, which consists of a coarse pseudo box generation module, a curriculum initialization module, and a prior and predictive collaborative curriculum refinement module.

As illustrated in Fig. 3, given a sequence of satellite video frames, a set of coarse pseudo boxes are first generated by spatio-guided matching, which leverages the location information from point annotations and the size information from proposal boxes extracted by an unsupervised background subtractor. In the curriculum initialization module, the initial priority of coarse pseudo boxes is captured by locally consistent prior, which incorporates three geometric measures: location deviation, scale variation, and pixel inconsistency. Throughout the training procedure of CFDRM, the collaborative curriculum refinement module updates the priority and refines the pseudo boxes by using prior and predictive knowledge. To alleviate the inherent ambiguity in the weak supervision paradigm, a novel ambiguity-aware loss is defined to optimize the coarse-to-fine dynamic refinement procedure.

### B. Coarse Pseudo Box Generation

Each point annotation indicates the existence of an object at a certain location, whereas it provides less cue on the object's contours. In this work, we adopt the motion foreground blobs extracted by an unsupervised background subtractor as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

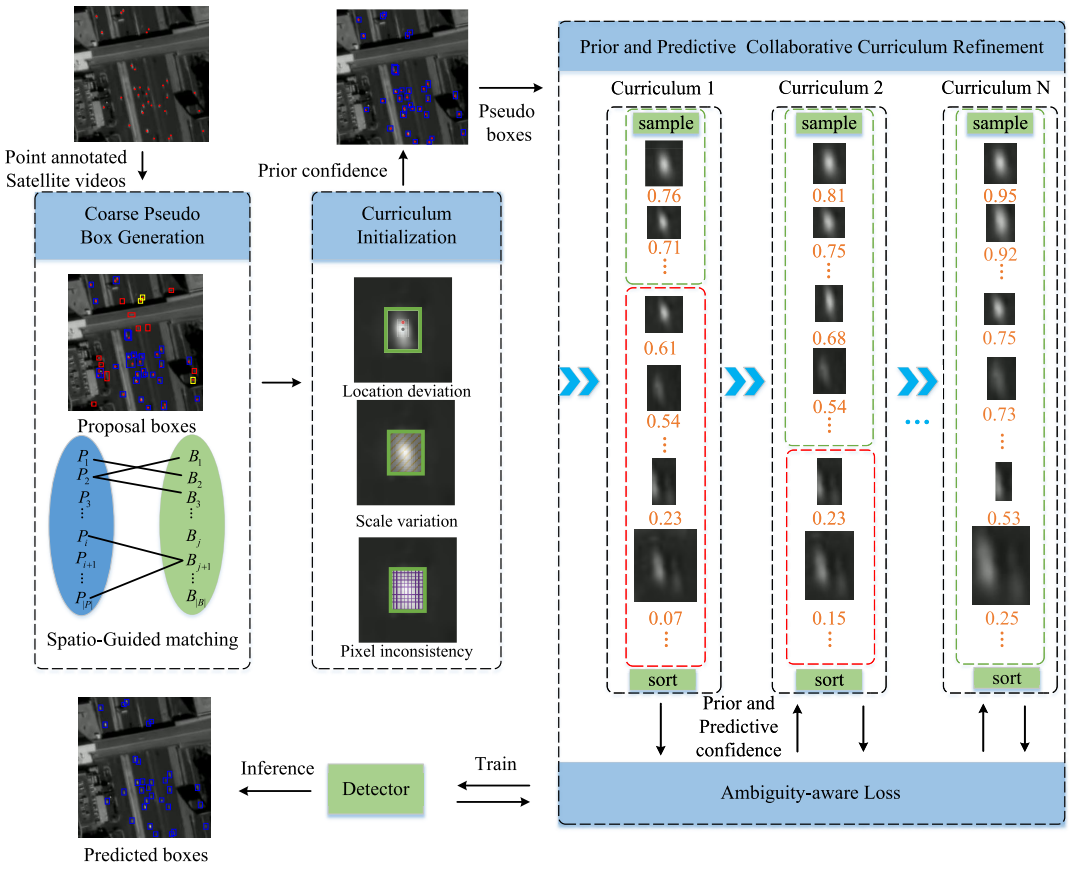



Fig. 3. Overview of CFDRM using only point-level annotations. It consists of a coarse pseudo box generation module, a curriculum initialization module, and a prior and predictive collaborative curriculum refinement module.

TABLE II

ABLATION EXPERIMENTS ON COARSE PSEUDO BOX GENERATION

| Matching | Merging | Splitting | Regenerating | Dubai | | San Diego | | Las Vegas | | VISO | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 |
| ✓ | | | | 69.19 | 30.28 | 65.23 | 30.24 | 56.12 | 25.04 | 51.77 | 13.45 |
| ✓ | ✓ | | | 69.87 | 35.02 | 66.34 | 34.25 | 59.34 | 33.26 | 54.76 | 28.82 |
| ✓ | ✓ | ✓ | | 70.73 | 40.21 | 67.12 | 39.19 | 63.25 | 38.05 | 55.23 | 30.42 |
| ✓ | ✓ | ✓ | ✓ | **72.12** | **50.01** | **69.07** | **48.11** | **66.25** | **44.19** | **58.54** | **35.31** |

TABLE III

ABLATION EXPERIMENTS ON PRIOR AND PREDICTIVE COLLABORATIVE CURRICULUM REFINEMENT

| Pred | Prior | Fixed | Adaptive | Dubai | | San Diego | | Las Vegas | | VISO | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 |
| ✓ | | ✓ | | 76.25 | 69.24 | 74.23 | 68.87 | 70.54 | 61.04 | 60.36 | 51.52 |
| ✓ | | | ✓ | 79.36 | 73.18 | 77.81 | 72.55 | 73.05 | 66.79 | 63.32 | 57.31 |
| ✓ | ✓ | ✓ | | 80.09 | 74.23 | 77.60 | 73.25 | 74.81 | 68.72 | 66.45 | 59.26 |
| ✓ | ✓ | | ✓ | **83.14** | **78.65** | **80.84** | **77.12** | **77.14** | **72.92** | **69.79** | **64.58** |

an extra cue on objects' contours. Due to the interference of background motion, the obtained foreground blobs are commonly of unsatisfactory precision, and directly extracting pseudo boxes from them commonly suffers from fragmented or merged boxes and missed objects. To opt for precise pseudo boundary boxes, we design a spatio-guided matching algorithm by analyzing the spatial relation between the foreground blobs and the point annotations, where the fragmented, merged, and missed objects are explicitly handled.

Given a video frame, let $\mathcal{P} = \{(x_i, y_i)|i = 1, 2, \ldots, |\mathcal{P}|\}$ denote the point annotations. We define the minimum bounding rectangle for each extracted foreground blob as a proposal. Note $\mathcal{B} = \{(x_j, y_j, w_j, h_j)|j = 1, 2, \ldots, |\mathcal{B}|\}$ as the set of proposal boxes corresponding to the extracted foreground blobs, and $|\mathcal{B}|$ refers to the number of the extracted foreground blobs. $(x_j, y_j)$ and $(w_j, h_j)$ are the center coordinate of the $j$th foreground blob and its width and height, respectively. A binary relation matrix $\mathbf{M} \in \{0, 1\}^{|\mathcal{P}|\times|\mathcal{B}|}$ is constructed by analyzing the spatial relation between the point annotations and foreground blobs, where $\mathbf{M}_{i,j} = 1$ indicates that point annotation $\mathcal{P}_i$ is inside the proposal box $\mathcal{B}_j$.

Inspired by the fact there exists a one-to-one assignment between point annotations and ideal foreground blobs, our pseudo box generation scheme starts by initializing the pseudo box set $\mathcal{S}$ by searching the pair of $(i, j)$ that satisfies $\sum_{k=1}^{|\mathcal{B}|} \mathbf{M}_{i,k} = 1$ and $\sum_{k=1}^{|\mathcal{P}|} \mathbf{M}_{k,j} = 1$. Due to the existence of fragmented, merged, and missed proposals, this procedure results in a set of unmatched point annotations, which are then fixed step-by-step.

*1) Handling Fragmented Proposals:* Multiple proposals extracted from fragmented foreground blobs may contain a single-point annotation, leading to an unexpected one-to-many assignment between point annotations and proposals, i.e., there exists a point $\mathcal{P}_i$ that satisfies $\sum_{k=1}^{|\mathcal{B}|} \mathbf{M}_{i,k} > 1$. For fixing this issue, a proposal with the smallest distance is selected as the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

assigned pseudo box to $\mathcal{P}_i$

$$\mathcal{B}_{\text{opt}}^{\min} = \arg\min_{\mathcal{B}_k \in \mathcal{B}^{\mathcal{P}_i}} d(\mathcal{B}_k, \mathcal{P}_i) \quad \text{s.t. } \mathcal{B}^{\mathcal{P}_i} = \left\{ \mathcal{B}_k \middle| \mathbf{M}_{i,k} = 1 \right\} \tag{1}$$

where $d(\cdot,\cdot)$ measures the Euclidean distances between a proposal's center and a point. $\mathcal{B}^{\mathcal{P}_i}$ gathers the proposal boxes containing the same point annotation $\mathcal{P}_i$. Then, the assigned $\mathcal{B}_{\text{opt}}^{\min}$ is added to our pseudo box set $\mathcal{S}$, while, for other element in $\mathcal{B}^{\mathcal{P}_i}$, the corresponding relation indicator in $\mathbf{M}$ is set to 0.

*2) Handling Merged Boxes:* Merged foreground blobs would cause multiple point annotations contained in a single proposal box, i.e., there exists a proposal box $\mathcal{B}_j$ satisfying $\sum_{k=1}^{|\mathcal{P}|} \mathbf{M}_{k,j} > 1$. In this case, for each point annotation, we create an individual pseudo boundary box and estimate its width and height on its minimal distance toward the box's vertical and horizontal boundaries

$$\begin{cases} w_i^{\text{half}} = \min\left( x_i^{\mathcal{P}} - \left( x_j^{\mathcal{B}} - \dfrac{w_j^{\mathcal{B}}}{2} \right), \left( x_j^{\mathcal{B}} + \dfrac{w_j^{\mathcal{B}}}{2} \right) - x_i^{\mathcal{P}} \right) \\ h_i^{\text{half}} = \min\left( y_i^{\mathcal{P}} - \left( y_j^{\mathcal{B}} - \dfrac{h_j^{\mathcal{B}}}{2} \right), \left( y_j^{\mathcal{B}} + \dfrac{h_j^{\mathcal{B}}}{2} \right) - y_i^{\mathcal{P}} \right) \end{cases} \tag{2}$$

where $w_i^{half}$ and $h_i^{half}$ are the minimum distances between each point annotation $\mathcal{P}_i$ in $\mathcal{P}^{\mathcal{B}_j}$ and the vertical and horizontal boundaries of the corresponding proposal box $\mathcal{B}_j$. Then, the estimated boxes are added to our pseudo box set $\mathcal{S}$.

*3) Handling Missed Objects:* The missed objects are not contained by any proposal box, i.e., there exists a point $\mathcal{P}_i$ with $\sum_{k=1}^{|\mathcal{B}|} \mathbf{M}_{i,k} = 0$. To ensure that these point annotations get reasonably pseudo boxes, we generate a new pseudo box for each of them based on statistics on obtained pseudo boxes

$$w_i^{\mathcal{P}}, h_i^{\mathcal{P}} \sim \mathcal{N}\left(\mu_w^{\mathcal{S}}, \sigma_w^{\mathcal{S}}\right), \mathcal{N}\left(\mu_h^{\mathcal{S}}, \sigma_h^{\mathcal{S}}\right) \tag{3}$$

in which the width and height are randomly sampled from Gaussian distributions with the means $\mu_w^{\mathcal{S}}$ and $\mu_h^{\mathcal{S}}$ and the standard deviations $\sigma_w^{\mathcal{S}}$ and $\sigma_h^{\mathcal{S}}$ estimated on elements from $\mathcal{S}$.

By incorporating these steps, we leverage the temporal information provided by the subtractor and the positional information derived from the point annotations. This allows us to generate an accurate pseudo boundary box, thereby improving the accuracy and robustness of vehicle detection in our model. The detailed procedure of the spatio-guided matching algorithm is listed in Algorithm 1. For the unsupervised extractor, we choose GMM due to its stability and effectiveness in capturing motion information, as presented in Table I.

**Algorithm 1** Spatio-Guided Matching

**Input:**
Point annotations $\mathcal{P} = \{(x_i, y_i) | i = 1, 2, \ldots, |\mathcal{P}|\}$ and proposal boxes $\mathcal{B} = \{(x_j, y_j, w_j, h_j) | j = 1, 2, \ldots, |\mathcal{B}|\}$;
**Output:**
Generated pseudo boundary boxes $\mathcal{S}$;

**Begin**
Initialize an empty set of pseudo boundary boxes $S$ and construct a binary relation matrix $\mathbf{M} \in \{0,1\}^{|\mathcal{P}| \times |\mathcal{B}|}$;
**for** each pair $(i, j)$ in $\mathbf{M}$: **do**
  **if** $\sum_{k=1}^{|\mathcal{B}|} \mathbf{M}_{i,k} = 1$ and $\sum_{k=1}^{|\mathcal{P}|} \mathbf{M}_{k,j} = 1$ **then**
    $\mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{B}_j$;
  **end if**
**end for**
**for** each row $i$ in $\mathbf{M}$: **do**
  **if** $\sum_{k=1}^{|B|} M_{i,k} > 1$ **then**
    $\mathcal{B}_i$ is obtained via (1);
    $\mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{B}_i$;
    Remove other redundant proposal boxes within $B^{\mathcal{P}_i}$;
  **end if**
**end for**
**for** each column $j$ in $\mathbf{M}$: **do**
  **if** $\sum_{k=1}^{|P|} M_{k,j} > 1$ **then**
    $\mathcal{B}_i$ is obtained via (2);
    $\mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{B}_i$;
  **end if**
**end for**
**for** each row $i$ in $\mathbf{M}$: **do**
  **if** $\sum_{k=1}^{|B|} M_{i,k} = 0$ **then**
    $\mathcal{B}_i$ is obtained via (3);
    $\mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{B}_i$;
  **end if**
**end for**
**return** $\mathcal{S}$.
**End**

## C. Curriculum Initialization

Previous single-point supervision methods commonly rely on empirical observations on object density for constructing the initial learning curriculum and underestimate the importance of prior information on pseudo boxes, leading to poor adaptability on satellite videos. To address this issue, we propose a novel locally consistent prior method that adaptively adjusts the sample difficulty by incorporating prior knowledge on the location deviation, scale variation, and pixel inconsistency of the obtained pseudo boxes.

*1) Location Deviation:* The metric $c_d^i$ measures the Euclidean distance between each pair of pseudo box's center and the corresponding point annotation. Pseudo boxes with larger spatial shifts from their assigned point annotations will be of higher difficulty.

*2) Scale Variation:* We assume that the scales of the generated pseudo boxes follow a Gaussian distribution and scale consistency is defined to penalized boxes with large-scale variation as:

$$c_a^i = \begin{cases} 0, & |\mu_a - a_i| \le 3\sigma_a \\ \log(1 + (|\mu_a - a_i| - 3\sigma_a)), & \text{otherwise} \end{cases} \tag{4}$$

where $a^i$ refers to the area of $i$th pseudo box, and $\mu_a$ and $\sigma_a$ are the mean and standard deviation estimated on all pseudo boxes in the current scene. More specifically, we treat pseudo boxes with scale in a certain range of $[\mu_a + 3\sigma_a, \mu_a - 3\sigma_a]$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION EXPERIMENTS ON LOCALLY CONSISTENT PRIOR

| $\alpha_d$ | $\alpha_a$ | $\alpha_p$ | Dubai | | San Diego | | Las Vegas | | VISO | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 |
| | | | 79.36 | 73.18 | 77.81 | 72.55 | 73.05 | 66.79 | 63.32 | 57.31 |
| ✓ | | | 81.34 | 74.76 | 78.65 | 74.93 | 75.45 | 68.02 | 65.54 | 59.46 |
| ✓ | ✓ | | 81.93 | 75.85 | 79.27 | 75.02 | 76.01 | 69.78 | 67.45 | 61.02 |
| ✓ | ✓ | ✓ | **83.14** | **78.65** | **80.84** | **77.12** | **77.14** | **72.92** | **69.79** | **64.58** |

as highlight confident samples, and their scale variations are forced to 0.

*3) Pixel Inconsistency:* We start by converting the RGB images into grayscale and adopt the variance $[(\sigma_{G_i})^2]$ of pixel values within the $i$th pseudo box as its pixel inconsistency measurement

$$c_p^i = \log\left(1 + \left(\sigma_{G_i}\right)^2\right) \tag{5}$$

in which the logarithmic operator is adopted for the transformation of the variance to monotonically increasing difficulty scores.

The overall difficulty of each sample summarizes its location deviation, scale variation, and pixel inconsistency, and then, it is transformed into a confident indicator by

$$\begin{aligned} c_{\text{prior}}^i &= \alpha_d c_d^i + \alpha_a c_a^i + \alpha_p c_p^i \\ w_{\text{prior}}^i &= e^{-c_{\text{prior}}^i} \end{aligned} \tag{6}$$

where $w_{\text{prior}}^i$ is the confidence of a pseudo box. $\alpha_d$, $\alpha_a$, and $\alpha_p$ are scalars that control the contribution of each penalty term. In our experiments, we set $\alpha_d$ to 0.5, $\alpha_a$ to 0.2, and $\alpha_p$ to 0.3.

Thus, for constructing the initial curriculum, the difficulty of the pseudo box is defined as the complement of its confidence score. We sort all the training samples and then split the overall training set $I$ into $Z$ folds $I_1, I_2, \ldots, I_Z$ accordingly. The training process begins with training the detector of CFDRM on the first fold, $I_1$, which contains pseudo boxes with the lowest difficulty. The training process continues by incorporating instances from the second fold, $I_2$, into the training set $(I_1 \cup I_2)$ and retraining the detector. Finally, this iterative procedure stops when $I_Z$ is included in the final curriculum.

### D. Prior and Predictive Collaborative Curriculum Refinement

In previous methods, the initial pseudo boxes are updated by selecting the predicted boxes with the highest scores by the detector. However, such a criterion ignores the prior information, resulting in inaccurate pseudo-box refinement and an unstable training process. To address these issues, we propose a prior and predictive collaborative curriculum refinement scheme, where prior information is considered explicitly.

*1) Box Refinement:* During each training epoch, the detector receives the pseudo boxes corresponding to their difficulty levels and performs forward propagation to generate predicted detection results and prediction scores.

The confidence of each pseudo box is then updated based on the predicted score and prior probability. For the $i$th pseudo box $\mathcal{S}_i^{(k)}$ in the $k$th epoch, based on point annotation information and non-maximum suppression (NMS) filtering, the most perfect matching predicted box $\hat{\mathcal{S}}_j^{(k)}$ is obtained. And the confidence score $\hat{c}_j^{(k)}$ of $\hat{\mathcal{S}}_j^{(k)}$ combines the prior and predictive information

$$c_j^{\hat{(k)}} = 0.5 \times s_j^{\hat{(k)}} + 0.5 \times w_{\text{prior}}^{j(k)} \tag{7}$$

where $s_j^{\hat{(k)}}$ and $w_{\text{prior}}^{j(k)}$ denotes the prediction score and prior score of $\hat{\mathcal{S}}_j^{(k)}$, respectively. In the initial training epoch, where no predicted score comparison is possible, we assign each pseudo box an initial confidence score $(c_i^1)$ equal to its prior probability $(w_{\text{prior}}^{i(1)})$.

If $\hat{c}_j^{(k)}$ exceeds the existing confidence score $c_i^{(k)}$ of $\mathcal{S}_i^{(k)}$, both the location and size of the pseudo box $\mathcal{S}_i^{(k+1)}$ in the $k$+1th epoch are refined with the corresponding prediction box $\hat{\mathcal{S}}_j^{(k)}$ and its confidence score is updated to $c_j^{\hat{(k)}}$. Conversely, $\mathcal{S}_i^{(k+1)}$ and $c_i^{(k+1)}$ remain the same as $\mathcal{S}_i^{(k)}$ and $c_i^{(k)}$, respectively.

*2) Curriculum Adjustment:* To ensure timely curriculum adjustments, following each training epoch, we recalculate the difficulty of those refined boxes generated during the box refinement procedure by considering the complement of their updated confidence scores. Following this, we reorder all pseudo boxes, divide them into $Z$ folds, and decide which folds of samples will be used as input for the detector based on the current training epoch. In our experiments, we have set $Z$ to 3, but this value can be adjusted according to specific requirements. And we employ CenterNet as the detector due to the loss design in Section II-E, but it can be replaced with other detectors as needed.

As summarized in Algorithm 2, our collaborative curriculum refinement scheme refines pseudo boxes and adjusts the curriculum, alternately. By following this curriculum-based strategy, the detector effectively learns from easier to more challenging samples, resulting in improved performance and enhanced adaptation to varying difficulty levels in object detection tasks.

### E. Ambiguity-Aware Loss

Instead of treating pseudo boxes uniformly, we propose a novel ambiguity-aware loss where the contributions of pseudo boundary boxes are dynamically adjusted based on their associated confidence scores, ensuring that boxes with higher confidence scores have enlarged impacts to the overall classification loss.

For an input frame in size of $W \times H$, we first produce a pseudo center keypoint heatmap $w_{xyc}^h \in [0, c_i^{(k)}]^{(W/R)\times(H/R)\times C}$, where $c_i^{(k)}$ is the confidence score of the $i$th pseudo box in the $k$th epoch, which is described in more detail in Section II-D, $R$ is the output stride, and $C$ is the classes. Since the object we detect is small, the larger the feature map, the better for detection, so $R$ is set to 2. Then, we put all pseudo center keypoints onto the heatmap $w_{xyc}^h$ using a Gaussian kernel $\exp(-(((x-\widetilde{p}_x)^2 + (y-\widetilde{p}_y)^2)/2\sigma_p^2))$ for each pseudo center keypoint $p \in R^2$ in which the low-resolution equivalent is $\widetilde{p} = \lfloor (p/R) \rfloor$, where $\sigma_p$ is an object size-adaptive standard deviation [20].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 2** Prior and Predictive Collaborative Curriculum Refinement

**Input:**

Initial pseudo boxes $\mathcal{S}^{(1)} = \{\mathcal{S}_i^{(1)} | i = 1, 2, \ldots, |\mathcal{S}^{(1)}|\}$, with their corresponding confidence scores $c_i^{(1)}$ which are set based on prior probabilities of the boxes $w_{\text{prior}}^{i(1)}$, $i = 1, 2, \ldots, |\mathcal{S}|$, the number of epochs $K$, the number of folds $Z$;

**Output:**

Refined pseudo boxes $\mathcal{S}^{(K)} = \{\mathcal{S}_i^{(K)} | i = 1, 2, \ldots, |\mathcal{S}^{(K)}|\}$;

**Begin**
**for** each epoch $k$ from 1 to $K$ **do**
  Sort $\mathcal{S}^{(k)}$ by instance difficulty;
  Split $\mathcal{S}^{(k)}$ into $Z$ folds, denoted as $I_1, I_2, \ldots, I_Z$;
  Train detector on $\{I_1 \cup I_2 \cup \cdots \cup I_{\lfloor \frac{k}{K} \rfloor \times Z}\}$;
  Obtain predicted boxes $\hat{\mathcal{S}^{(k)}}$ and prediction score $\hat{s^{(k)}}$;
  **for** $\mathcal{S}_i^{(k)}$ in $\mathcal{S}^{(k)}$ **do**
    Match the optimal predicted box $\hat{\mathcal{S}_j^{(k)}}$;
    Calculate $c_j^{(k)}$ by Eq.7;
    **if** $c_j^{(k)} > c_i^{(k)}$ **then**
      Update the location and size: $\mathcal{S}_i^{(k+1)} \Leftarrow \hat{\mathcal{S}_j^{(k)}}$;
      Update the confidence: $c_i^{(k+1)} \Leftarrow \hat{c_j^{(k)}}$;
    **else**
      Remain unchanged: $\mathcal{S}_i^{(k+1)} \Leftarrow \mathcal{S}_i^{(k)}$;
      Remain unchanged: $c_i^{(k+1)} \Leftarrow c_i^{(k)}$;
    **end if**
  **end for**
**end for**
**return** $\mathcal{S}^{(K)}$.
**End**

In addition, we produce another pseudo box heatmap $w_{xyc}^b \in [1, \exp(c_i^{(k)})]^{(W/R)\times(H/R)\times C}$, which means the weight of the $i$th box in an image is $b_{i(x_1:x_2,y_1:y_2)} = \exp(c_i^{(k)})$ and others are 1.

After obtaining two heatmaps, namely, $w_{xyc}^h$ and $w_{xyc}^b$, we incorporate them in the classification loss $\mathcal{L}_k$ within the CenterNet detector [21], which can be expressed as follows:

$$\mathcal{L}_{\text{amb}} = \begin{cases} w_{xyc}^h w_{xyc}^b, & \text{if } Y_{xyc} = 1 \\ \left(1 - w_{xyc}^h\right) w_{xyc}^b, & \text{otherwise} \end{cases}$$
$$\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{amb}} \mathcal{L}_k \tag{8}$$

where $Y_{xyc} = 1$ denotes the pseudo label of pixel $(x, y)$ of class $c$, and $\mathcal{L}_{\text{cls}}$ represents the newly weighted classification loss. And the overall loss of CFDRM, following CenterNet, is calculated as follows:

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{size}} \mathcal{L}_{\text{size}} + \lambda_{\text{off}} \mathcal{L}_{\text{off}} \tag{9}$$

where $\lambda_{\text{size}}$ and $\lambda_{\text{off}}$ are the two hyperparameters to balance different terms. In this article, $\lambda_{\text{size}}$ is set to be 0.1 and $\lambda_{\text{off}}$ is set to be 1.

It is important to note that due to the curriculum adjustment procedure outlined in Section II-D, we only compute the loss for a subset of pseudo boxes within the image. Our loss formulation is general and can be applied to various weakly supervised object detection frameworks. Enhanced by the ambiguity-aware loss, our CFDRM can be efficiently optimized to generate pseudo boxes.

TABLE V
ABLATION EXPERIMENTS ON AMBIGUITY-AWARE LOSS AND CL

| $w_{xyc}^h$ | $w_{xyc}^b$ | CL | Dubai | | San Diego | | Las Vegas | | VISO | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 | AP30 | AP50 |
| | | | 83.14 | 78.65 | 80.84 | 77.12 | 77.14 | 72.92 | 69.79 | 64.58 |
| ✓ | | | 84.32 | 79.67 | 83.51 | 78.67 | 79.03 | 74.86 | 71.02 | 64.91 |
| | ✓ | | 84.11 | 79.61 | 82.67 | 78.06 | 78.27 | 73.47 | 70.45 | 64.37 |
| ✓ | ✓ | | 85.14 | 80.15 | 83.80 | 79.58 | 80.14 | 75.62 | 72.70 | 65.33 |
| ✓ | ✓ | ✓ | **85.35** | **80.24** | **83.83** | **80.09** | **80.19** | **76.58** | **73.45** | **67.19** |

TABLE VI
DETECTION RESULTS OF DIFFERENT ANNOTATIONS

| (%) ↑ | Dubai | | | San Diego | | | Las Vegas | | | VISO | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pr | Re | F1 | Pr | Re | F1 | Pr | Re | F1 | Pr | Re | F1 |
| **CFDRM (Coarse)** | 81.15 | 87.67 | 84.28 | 79.11 | 85.69 | 82.27 | 73.20 | 83.02 | 77.80 | 68.25 | 81.42 | 74.26 |
| **CFDRM (Fine)** | 82.11 | 88.25 | 85.07 | 80.04 | 86.16 | 82.99 | 75.36 | 84.89 | 79.84 | 70.26 | 83.72 | 76.40 |
| **CFDRM (Box)** | 86.75 | 89.74 | 88.22 | 82.14 | 88.87 | 85.37 | 78.16 | 87.26 | 82.46 | 72.46 | 85.74 | 78.54 |

## III. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Dataset

In this article, four challenging satellite video datasets for moving vehicle detection are adopted for performance evaluation, as shown in Fig. 4. The details of these datasets are summarized as follows.

*1) Dubai Dataset:* This dataset contains three satellite videos captured over Dubai, UAE, on November 9, 2018. Each of them contains 320 frames with a frame rate of 10 frames/s. Each video frame is in size of 1000 × 1000 pixels, and the spatial resolution is 0.92 m per pixel. We used two of them for training and the remaining one for testing.

*2) San Diego Dataset:* This dataset is constructed from a satellite video captured over San Diego, CA, USA, on May 23, 2017. It contains two satellite videos in size of 1000 × 1000 pixels and one satellite video in size of 1500 × 700 pixels. These videos share the same frame rate and spatial resolution as those from Dubai dataset. During our experiment, two of these videos are selected for training, while the remaining one is utilized for testing.

*3) Las Vegas Dataset:* Collected over Las Vegas, NV, USA, on March 25, 2014, this dataset has two satellite videos, both containing 700 frames. Among these videos, one is in size of 600 × 400 pixels and used for training, and the other one is in size of 400 × 400 pixels and used for testing.

*4) VISO Dataset:* The VISO [45] dataset is composed of 77 satellite videos captured by the Jilin-1 satellite constellation at different positions of the satellite orbit. Each video consists of approximately 350 frames at 10 frames/s and their spatial resolution is 0.92 m per pixel. There are 70 videos as the training set and seven videos as the test set.

In these datasets, the objects are annotated with axis-aligned boundary boxes. In this work, we employed a random function to randomly select a pixel point within the box as the point annotation.

### B. Implementation Details

The moving vehicles in the videos were selected as the targets. In the label generations, we have tried several traditional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

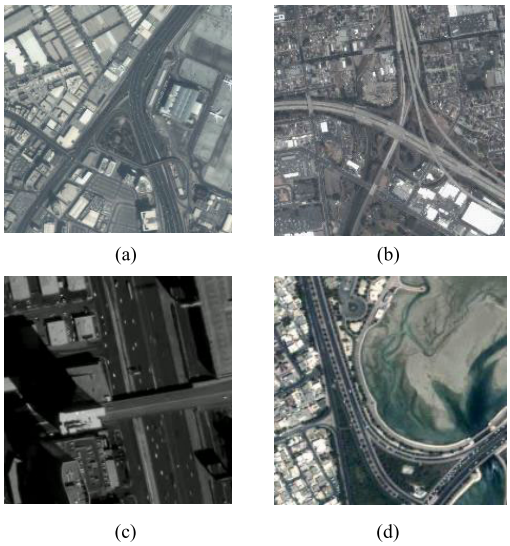


Fig. 4. Exemplar frames from selected satellite video datasets: (a), (b), and (d) are from Dubai, San Diego, and VISO datasets captured by the Jilin-1 satellite, respectively, and (c) is from the Las Vegas dataset captured by the SkySat satellite.

TABLE VII
PARAMETER ANALYSIS OF THE WEIGHT OF LOCALLY CONSISTENT PRIOR ON THE DUBAI DATASET

| $\alpha_d$ | $\alpha_a$ | $\alpha_p$ | Prec(%) ↑ | Rec (%) ↑ | F1(%) ↑ |
|---|---|---|---|---|---|
| 0.5 | 0.2 | 0.3 | **81.15** | **87.67** | **84.28** |
| 0.4 | 0.2 | 0.3 | 81.07 | 87.41 | 84.12 |
| 0.6 | 0.2 | 0.3 | 81.05 | 87.54 | 84.17 |
| 0.5 | 0.3 | 0.3 | 80.87 | 86.50 | 83.59 |
| 0.5 | 0.1 | 0.3 | 81.02 | 87.30 | 84.04 |
| 0.5 | 0.2 | 0.4 | 81.10 | 87.64 | 84.24 |
| 0.5 | 0.2 | 0.2 | 81.05 | 87.45 | 84.13 |

TABLE VIII
SENSITIVITY ANALYSIS OF GMM WITH DIFFERENT FRAME RATES

| | F1(%) ↑ | | | |
|---|---|---|---|---|
| $\tau$ | Dubai | San Diego | Las Vegas | VISO |
| 1 | 50.80 | 48.75 | 50.29 | 14.67 |
| 3 | 61.09 | 59.56 | 56.80 | 35.21 |
| 5 | **62.08** | **60.15** | **57.06** | **36.34** |
| 7 | 60.25 | 59.24 | 56.23 | 35.88 |

unsupervised methods, and the performance of GMM has been shown to be robust to the initialization results, as can be seen in Table I. During the training phase, the Adam optimizer is adopted with an initial learning rate of $3.2 \times 10^{-4}$, and the learning rate is decreased by a factor of 0.1 at the 30th and 50th epochs, respectively. We train our models for 65 epochs with a batch size of 4 on an NVIDIA RTX 3090 GPU. The training data are randomly cropped as $256 \times 256$ pixels per frame, and random flipping and rotation processes are adopted as data augmentations. During inference, the cropped original frames are fed to the model without extra data augmentations.

### C. Experimental Results and Comparisons

To demonstrate the superior performance of our CFDRM, we compare its detection performance against a variety of moving object detectors in unsupervised, weakly supervised, and fully supervised manners on the selected datasets. Specifically, some representative unsupervised methods, FastMCD, Vibe, and GMM, existing weakly supervised methods, LSC-CNN, P2BNet, and PLUG, and fully supervised methods, Faster R-CNN, Cascade R-CNN, CornerNet, CenterNet, RepPoints, CentripetalNet, and DSFNet, are selected for comparison. For a fair comparison, Faster R-CNN, Cascade R-CNN, CornerNet, CenterNet, RepPoints, CentripetalNet, and DSFNet take three stacked frames as the input to exploit the temporal information. We use the common evaluation of precision, recall, and F1-score for all the experiments. Given the small size of objects in satellite videos, the intersection over union (IoU) threshold for correct detection is set to 0.3.

As summarized in Table I, the CFDRM model has achieved the state-of-the-art results on four datasets when compared to weakly supervised methods. The achieved F1-scores by our CFDRM models were 84.28%, 82.27%, 77.80%, and 74.26% on our selected datasets, outperforming the existing best-performing unsupervised method by significant margins of 22.2%, 22.12%, 20.74%, and 37.92%, respectively. At the same time, our CFDRM surpasses the existing best-performing weakly supervised method P2BNet by significant margins of 12.87%, 11.56%, 14.95%, and 34.43%, respectively. Compared with existing fully supervised methods, our approach achieves competitive results and in some cases surpasses our baseline CenterNet. We also constructed another set of experiments where the boundary boxes' centers are directly adopted as point annotations. Utilizing the center point as the point annotations indicates the robustness of CFDRM to variations in point annotation positions. As shown in Table I, the shift from center point annotations (Fine) to noncenter point annotations (Coarse) results in the smallest performance decline for CFDRM when compared to other weakly supervised point-based methods. In addition, the noncenter point annotations are better aligned with real-world labeling scenarios.

On the other hand, the performance gain by our CFDRM can also be validated qualitatively. The unsupervised methods, Vibe and FastMCD, cause a large number of false negatives for slowly moving objects, especially in the Dubai and San Diego datasets, which can be seen in Figs. 5 and 6(b) and (c). This is due to their insufficient utilization of motion information. Although GMM is relatively effective in extracting motion information, it is difficult to avoid the false alarms in the background region caused by brightness and contrast changes, especially for blurry and dense vehicles. For existing weakly supervision methods, P2BNet and PLUG generate a large number of false alarms, and missing detection and a large portion of the generated boundary boxes are enlarged, as presented in Figs. 5–8. This performance degradation is primarily attributed to the absence of motion information, resulting in imprecise initialization and restricted refinement during subsequent training processes. Compared with the unsupervised and weakly supervised methods, fully supervised methods improve the detection performance significantly. Our CFDRM is nearly indistinguishable from these fully supervised methods, which validates the superiority of our CFDRM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

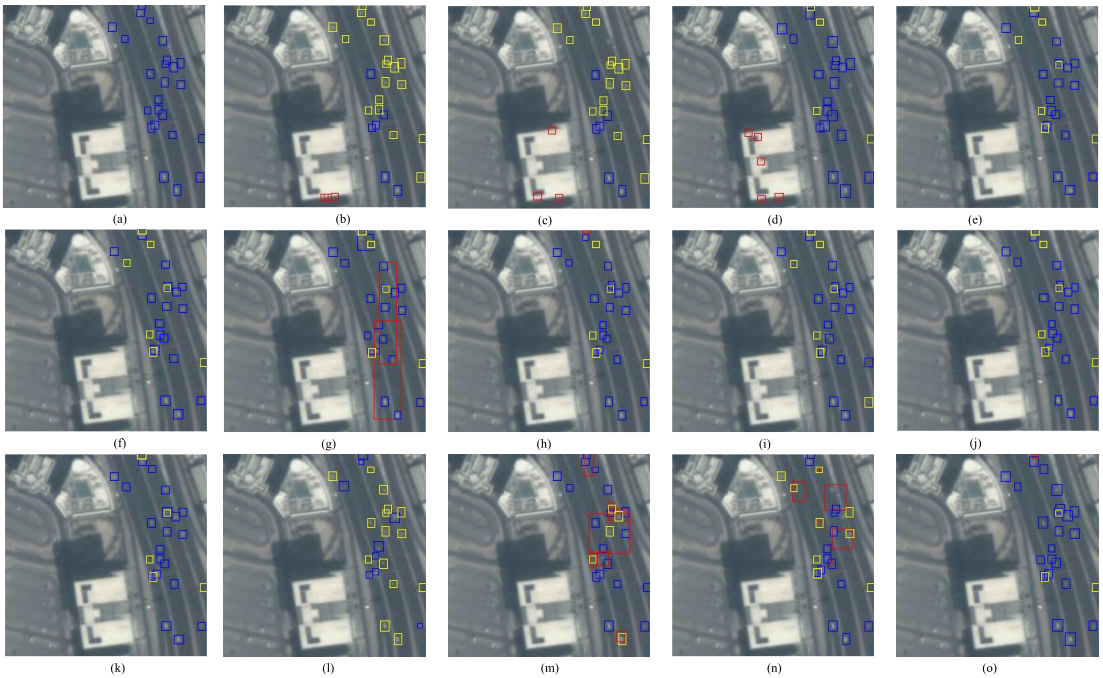


Fig. 5. Here are the detection results from the Dubai satellite video dataset: true positives are highlighted in blue, false negatives are indicated in yellow, and false positives are shown in red. (a) Ground truth. (b) Vibe. (c) FastMCD. (d) GMM. (e) Faster R-CNN. (f) Cascade R-CNN. (g) CornerNet. (h) CenterNet. (i) RepPoints. (j) CentripetalNet. (k) DSFNet. (l) LSC-CNN. (m) P2BNet. (n) PLUG. (o) CFDRM.

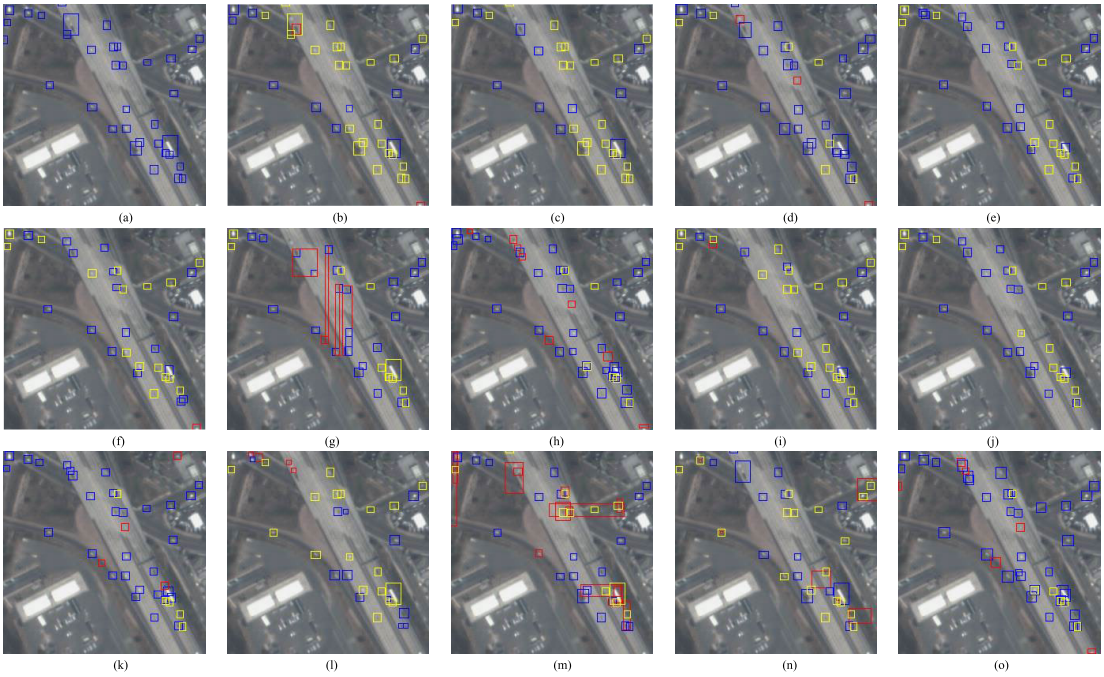


Fig. 6. Here are the detection results from the San Diego satellite video dataset: true positives are highlighted in blue, false negatives are indicated in yellow, and false positives are shown in red. (a) Ground truth. (b) Vibe. (c) FastMCD. (d) GMM. (e) Faster R-CNN. (f) Cascade R-CNN. (g) CornerNet. (h) CenterNet. (i) RepPoints. (j) CentripetalNet. (k) DSFNet. (l) LSC-CNN. (m) P2BNet. (n) PLUG. (o) CFDRM.

### *D. Ablation Study*

In the proposed CFDRM, a coarse pseudo box generation module and a prior and predictive collaborative curriculum refinement module are developed, and the locally consistent prior and ambiguity-aware loss are adopted to further eliminate the uncertainty caused by weak supervision. In this section, we present a series of ablation studies to examine their contributions to the overall performance gain by CFDRM. To more effectively validate the impact of each module on object location accuracy, we utilize both $AP_{50}$ and $AP_{30}$ as metrics. Specifically, $AP_{50}$ and $AP_{30}$ indicate the percentage of correct detections by the model when the overlap between the predicted bounding box and the actual bounding box exceeds 50% and 30%, respectively. A high $AP_{50}$ score indicates that the model is very accurate in locating objects closely to their true positions. A high $AP_{30}$ score suggests that the model is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

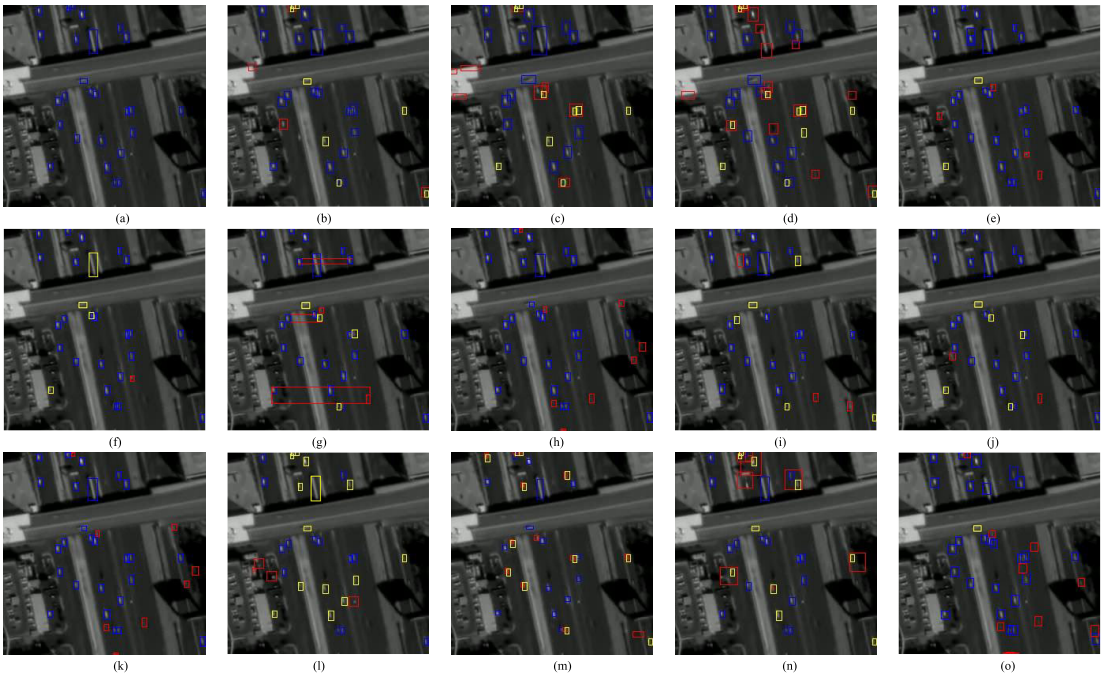



Fig. 7. Here are the detection results from the Las Vegas satellite video dataset: true positives are highlighted in blue, false negatives are indicated in yellow, and false positives are shown in red. (a) Ground truth. (b) Vibe. (c) FastMCD. (d) GMM. (e) Faster R-CNN. (f) Cascade R-CNN. (g) CornerNet. (h) CenterNet. (i) RepPoints. (j) CentripetalNet. (k) DSFNet. (l) LSC-CNN. (m) P2BNet. (n) PLUG. (o) CFDRM.

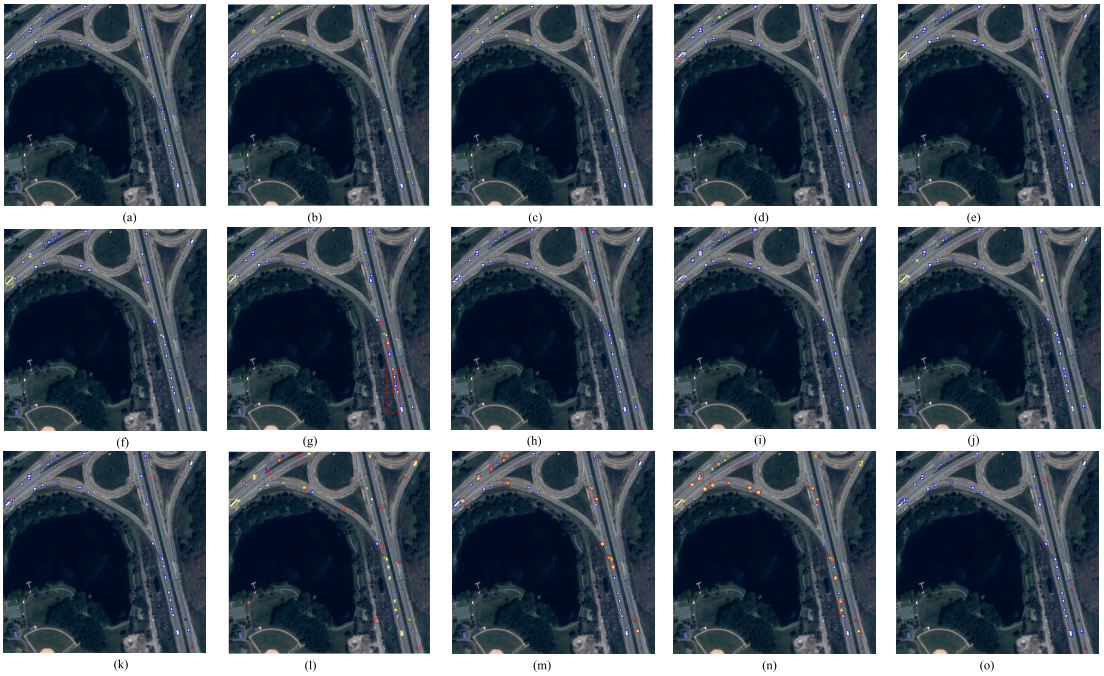



Fig. 8. Here are the detection results from the VISO satellite video dataset: true positives are highlighted in blue, false negatives are indicated in yellow, and false positives are shown in red. (a) Ground truth. (b) Vibe. (c) FastMCD. (d) GMM. (e) Faster R-CNN. (f) Cascade R-CNN. (g) CornerNet. (h) CenterNet. (i) RepPoints. (j) CentripetalNet. (k) DSFNet. (l) LSC-CNN. (m) P2BNet. (n) PLUG. (o) CFDRM.

generally good at detecting the presence of objects, even if the precision of the exact location is less strict.

*1) Ablation on Coarse Pseudo Box Generation:* Spatio-guided matching algorithm opts for more precise pseudo boundary boxes from point annotations and moving foreground blobs. For handling inaccurate foreground blobs caused by an unsupervised background subtraction method, we design a series of operations. In this section, we conducted a series of ablations for examining their effects on the overall performance gain, as summarized in Table II.

In the baseline model, only matching is applied. When merging is introduced for handling fragmented blobs, the detection precision is improved by 15.37% and 2.99% on VISO dataset in terms of the $AP_{50}$ and $AP_{30}$, respectively. Similar improvements are also observed on Dubai, San Diego, and Las Vegas datasets. After applying splitting for handling

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

merged boxes, the performance metrics are further improved by 4.79% and 3.91% in terms of $AP_{50}$ and $AP_{30}$ in the Las Vegas dataset, respectively. When all operations are used, the $AP_{50}$ and $AP_{30}$ achieve their highest values of 35.31% and 58.54% in the VISO dataset, respectively.

*2) Ablation on Prior and Predictive Collaborative Curriculum Refinement:* Prior and predictive collaborative curriculum refinement is proposed as an iterative approach to refine the pseudo boxes during the training process, aiming to address the issue of inaccurate pseudo boxes that can lead to decreased accuracy. In this work, we conducted experiments using different combinations of strategies, as summarized in Table III.

In the baseline model, a fixed confidence threshold was employed for updating the pseudo boxes. However, by replacing the conventional fixed threshold with our adaptive strategy, we observed consistent improvements in both $AP_{50}$ and $AP_{30}$ across all datasets, with an average increase of 3%–5%. The addition of prior confidence further enhanced the performance, resulting in an average improvement of 4%–6% compared to the scenario without prior confidence adjustment. The combination of the adaptive strategy and prior confidence adjustment yielded the best results, achieving an $AP_{50}$ of 72.92% and an $AP_{30}$ of 77.14% in the Las Vegas dataset. These findings highlight the effectiveness of our proposed adaptive strategy and the importance of incorporating prior confidence to refine pseudo boundary boxes.

*3) Ablation on Locally Consistent Prior:* Table IV presents the results of our ablation experiments on the locally consistent prior. We employed three geometric measures, namely, location deviation, scale variation, and pixel inconsistency, to capture the prior confidence of pseudo boxes.

Analyzing Table IV, we observe that the incorporation of different dimensions of prior information leads to incremental improvements in both $AP_{50}$ and $AP_{30}$. Specifically, when considering location deviation alone, we observe an average improvement of 2% in terms of $AP_{50}$ and $AP_{30}$ across all datasets compared to the baseline model. Moreover, the inclusion of scale variation further enhances the $AP_{50}$ score by an additional 1.76% in the Las Vegas dataset. When all three geometric measures are considered, we achieve the highest values of 64.58% and 69.79% in $AP_{50}$ and $AP_{30}$, respectively, in the VISO dataset.

*4) Ablation on Ambiguity-Aware Loss and CL:* In Table V, for the ambiguity-aware loss in lines 2–4, we can find that the confidence level is different from that of the supervised learning label, which is 1. If we consider the confidence level of the pseudo box and apply it to loss, the final result will be greatly improved, achieving a value of nearly 2%–3% on each dataset.

With the addition of CL strategies in the fifth line, we found that the performance of VISO with larger datasets increased by 1.86%, while the improvement effect of other smaller datasets was not obvious. This may be due to the fact that there are more instances of VISO dataset, and the gap of samples of different difficulty is large, while the problem does not exist in other smaller datasets.

*5) Ablation on Different Annotations:* The accuracy and detail of annotations significantly influence the performance of CFDRM. In our experiment, we categorized the types of annotations as follows: Coarse denotes the arbitrary point annotations, Fine indicates the center point annotations, and Box refers to bounding box annotations. In Table VI, the data from the first two lines indicate that the precision of the point annotation position plays a crucial role in the performance of the CFDRM model. Notably, this influence is more significant in the Las Vegas and VISO datasets, which see an approximate 2% increase in the F1-score, whereas the other datasets exhibit only a 1% improvement.

Furthermore, when comparing point and box annotations, there is a clear boost in performance across all datasets, with an evident 3% improvement when using box annotations. This enhancement highlights the benefits of using more detailed and comprehensive annotation information, which significantly aids the network's learning process.

### E. Parameter Analysis

*1) Parameter Analysis of the Weight of Locally Consistent Prior:* In CFDRM, the locally consistent prior is not only used for initial pseudo boxes but also used for subsequent updates to the pseudo boxes. CFDRM with different weights of locally consistent prior is analyzed, and the experimental results are shown in Table VII. The parameter $\alpha_d$ is set from 0.4 to 0.6 with the interval of 0.1, the parameter $\alpha_a$ is set from 0.1 to 0.3 with the interval of 0.1 and the parameter $\alpha_p$ is set from 0.2 to 0.4 with the interval of 0.1. CFDRM achieves the highest F1-score when $\alpha_d = 0.5$, $\alpha_a = 0.2$, and $\alpha_p = 0.3$.

*2) Sensitivity Analysis of GMM With Different Frame Rates:* The input frame rate determines the speed of the moving objects between frames, which affects the quality of the generated proposal boxes by GMM. In the experiment, $\tau$ represents the input sampling interval of the original frame. We changed $\tau$ to explore its impact on GMM. With a smaller $\tau$, the displacement of vehicles between frames may be too weak to provide explicit motion information for detection. When $\tau$ is too large, the vehicles move violently and the background is difficult to align, which hinders GMM from extracting stable motion information. As shown in Table VIII, GMM gets the best detection performance with $\tau = 5$.

## IV. Conclusion

In this article, a novel CFDRM is proposed for moving vehicle detection in satellite videos with only point-level annotations. The CFDRM divides the prediction of pseudo boxes into initialization and refinement stages. In the initialization stage, the most coarse proposal boxes are predicted through the off-the-shelf proposal method GMM, and adaptive sampling matching and adjustment is performed around the annotated points by spatio-guided matching. The refinement stage performs adaptive adjustment around the pseudo boxes to iteratively fine-tune the predicted boxes every epoch in a cascaded fashion. By combining these features, our CFDRM achieves promising results on our challenge datasets. In the future, we will further extend our method to generate rotated pseudo boxes from single-point labels and investigate more stable efficient pseudo label generation schemes. We hope

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

this study can draw attention to the research of single-point supervised vehicle detection in satellite videos.

## ACKNOWLEDGMENT

Our source code is available at https://github.com/jiefeng0109/CFDRM.

## REFERENCES

[1] X. Wang, K. Tan, P. Du, B. Han, and J. Ding, "A capsule-vectored neural network for hyperspectral image classification," *Knowl.-Based Syst.*, vol. 268, May 2023, Art. no. 110482.

[2] R. Ji, K. Tan, X. Wang, C. Pan, and L. Xin, "PASSNet: A spatial–spectral feature extraction network with patch attention module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[3] X. Wang, K. Tan, P. Du, C. Pan, and J. Ding, "A unified multiscale learning framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4508319.

[4] X. Wang et al., "Double U-Net (W-Net): A change detection network with two heads for remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103456.

[5] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[6] W. Ao, Y. Fu, X. Hou, and F. Xu, "Needles in a haystack: Tracking city-scale moving vehicles from continuously moving satellite," *IEEE Trans. Image Process.*, vol. 29, pp. 1944–1957, 2020.

[7] G. Kopsiaftis and K. Karantzalos, "Vehicle detection and traffic density monitoring from very high resolution satellite video data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1881–1884.

[8] J. Zhang, X. Jia, J. Hu, and K. Tan, "Moving vehicle detection for remote sensing video surveillance with nonstationary satellite platform," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5185–5198, Sep. 2022.

[9] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," 2023, *arXiv:2311.07113*.

[10] Y. Li, D. Hong, C. Li, J. Yao, and J. Chanussot, "HD-Net: High-resolution decoupled network for building footprint extraction via deeply supervised body and boundary decomposition," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 51–65, Mar. 2024.

[11] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[12] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.

[13] O. Barnich and M. Van Droogenbroeck, "ViBe: A universal background subtraction algorithm for video sequences," *IEEE Trans. Image Process.*, vol. 20, no. 6, pp. 1709–1724, Jun. 2011.

[14] Z. Zivkovic, "Improved adaptive Gaussian mixture model for background subtraction," in *Proc. 17th Int. Conf. Pattern Recognit.*, 2004, pp. 28–31.

[15] J. Zhang, X. Jia, J. Hu, and J. Chanussot, "Online structured sparsity-based moving-object detection from satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6420–6433, Sep. 2020.

[16] A. Abdelli and H.-J. Choi, "A four-frames differencing technique for moving objects detection in wide area surveillance," in *Proc. IEEE Int. Conf. Big Data Smart Comput.*, Feb. 2017, pp. 210–214.

[17] Jianguo Ju and Jinsheng Xing, "Moving object detection based on smoothing three frame difference method fused with rpca," *Multimedia tools Appl.*, vol. 78, pp. 29937–29951, May 2019.

[18] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1–18.

[19] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[20] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[21] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[22] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9657–9666.

[23] Z. Dong, G. Li, Y. Liao, F. Wang, P. Ren, and C. Qian, "CentripetalNet: Pursuing high-quality keypoint pairs for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10519–10528.

[24] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[25] R. LaLonde, D. Zhang, and M. Shah, "ClusterNet: Detecting small objects in large scenes by exploiting spatio-temporal information," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4003–4012.

[26] C. Xiao et al., "DSFNet: Dynamic and static fusion network for moving object detection in satellite videos," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] J. Feng, Y. Liang, X. Zhang, J. Zhang, and L. Jiao, "SDANet: Semantic-embedded density adaptive network for moving vehicle detection in satellite videos," *IEEE Trans. Image Process.*, vol. 32, pp. 1788–1801, 2023.

[28] H. Bilen and A. Vedaldi, "Weakly supervised deep detection networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2846–2854.

[29] P. Tang, X. Wang, X. Bai, and W. Liu, "Multiple instance detection network with online instance classifier refinement," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2843–2851.

[30] P. Tang et al., "PCL: Proposal cluster learning for weakly supervised object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 1, pp. 176–191, Jan. 2020.

[31] Z. Chen, Z. Fu, R. Jiang, Y. Chen, and X.-S. Hua, "SLV: Spatial likelihood voting for weakly supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12992–13001.

[32] D. P. Papadopoulos, J. R. R. Uijlings, F. Keller, and V. Ferrari, "Training object class detectors with click supervision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 180–189.

[33] Y. Liu, M. Shi, Q. Zhao, and X. Wang, "Point in, box out: Beyond counting persons in crowds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 6469–6478.

[34] D. B. Sam, S. V. Peri, M. N. Sundararaman, A. Kamath, and R. V. Babu, "Locate, size, and count: Accurately resolving people in dense crowds via detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2739–2751, Aug. 2021.

[35] Z. Ren, Z. Yu, X. Yang, M.-Y. Liu, A. G. Schwing, and J. Kautz, "UFO$^2$: A unified framework towards omni-supervised object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 288–313.

[36] Y. Wang, J. Hou, X. Hou, and L.-P. Chau, "A self-training approach for point-supervised object detection and counting in crowds," *IEEE Trans. Image Process.*, vol. 30, pp. 2876–2887, 2021.

[37] P. Chen et al., "Point-to-box network for accurate object detection via single point supervision," in *Proc. Eur. Conf. Comput. Vis.*, vol. 13669, 2022, pp. 51–67.

[38] S. He, H. Zou, Y. Wang, B. Li, X. Cao, and N. Jing, "Learning remote sensing object detection with single point supervision," 2023, *arXiv:2305.14141*.

[39] K. E. Van de Sande, J. R. Uijlings, T. Gevers, and A. W. Smeulders, "Segmentation as selective search for object recognition," in *Proc. Int. Conf. Comput. Vis.*, 2011, pp. 1879–1886.

[40] C. L. Zitnick and P. Dollár, "Edge boxes: Locating object proposals from edges," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 391–405.

[41] P. Arbeláez, J. Pont-Tuset, J. Barron, F. Marques, and J. Malik, "Multiscale combinatorial grouping," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 328–335, doi: 10.1109/CVPR.2014.49.

[42] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, "Single-image crowd counting via multi-column convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, May 2016, pp. 589–597.

[43] X. Yu et al., "Object localization under single coarse point supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4868–4877.

[44] K. Moo Yi, K. Yun, S. Wan Kim, H. Jin Chang, H. Jeong, and J. Young Choi, "Detection of moving objects with non-stationary cameras in 5.8ms: Bringing motion detection to your mobile device," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2013, pp. 27–34.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.

[45] Q. Yin et al., "Detecting and tracking small and dense moving objects in satellite videos: A benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612518.



**Jie Feng** (Senior Member, IEEE) received the B.S. degree from Chang'an University, Xi'an, China, in 2008, and the Ph.D. degree from Xidian University, Xi'an, in 2014.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Xidian University. Her research interests include remote sensing image processing, deep learning, and machine learning.



**Quanpeng Jiang** received the B.S. degree from the Dongbei University of Finance and Economics, Dalian, China, in 2021. He is currently pursuing the M.S. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include deep learning and remote sensing image processing.



**Junpeng Zhang** (Senior Member, IEEE) received the B.Sci. degree from the China University of Mining and Technology, Xuzhou, China, in 2013, the master's degree in surveying engineering from the China University of Mining and Technology, in 2016, and the Ph.D. degree in electrical engineering from The University of New South Wales, Australia, in 2021.

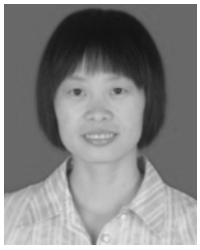
He serves as an Associated Professor with Xidian University. His research interests include object detection and tracking in remote sensing imaginary.

Dr. Zhang was the winner of "DSTG Best Contribution to Science Award" in Digital Image Computing: Techniques and Applications 2018 (DICTA 2018). He served as the chair of IEEE GRSS UNSW Canberra Student Chapter during 2020.



**Yuping Liang** received the B.S. and M.S. degrees from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2019 and 2022, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include deep learning and remote sensing data analysis and understanding.



**Ronghua Shang** (Senior Member, IEEE) received the B.S. degree in information and computation science and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with Xidian University. Her research interests include machine learning, pattern recognition evolutionary computation, image processing, and data mining.



**Licheng Jiao** (Fellow, IEEE) received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

He has authored or coauthored more than 150 scientific articles. He has charged of about 40 important scientific research projects and authored more than 20 monographs and 100 articles in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:15:40 UTC from IEEE Xplore. Restrictions apply.